PHISHSCANQR

MOBILE DETECTION OF PHISHING QR CODE USING VIRUSTOTAL AND MACHINE LEARNING

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ABSTRACT

The rise of QR codes in various sectors has inadvertently introduced a new cyber threat known as "quishing," or QR code phishing, which obscures malicious URLs within legitimate-looking codes. A notable example involves attackers placing fake QR codes on parking meters to redirect victims to fraudulent payment sites for credit card information theft. To combat this growing threat, a mobile detection system is proposed that employs machine learning and integrates with the VirusTotal API to evaluate QR code URLs. A supervised classifier is utilized to identify suspicious patterns while concurrently checking URLs against VirusTotal's threat database, enhancing detection accuracy and minimizing false positives. The Android application provides real-time alerts and visual indicators, empowering users to make informed decisions before interacting with potentially harmful links, thereby bolstering mobile security against QR code phishing.

PROBLEM STATEMENT

- Traditional email security solutions focus on text and URL analysis, which often fail to detect malicious content in images like QR codes.
- Phishing detection methods primarily utilize URL-based classifiers that depend on incomplete and slow-to-update phishing databases.
- Existing email security filters and intrusion detection systems (IDS) are typically ineffective in decrypting QR codes and scanning inner URLs for threats.

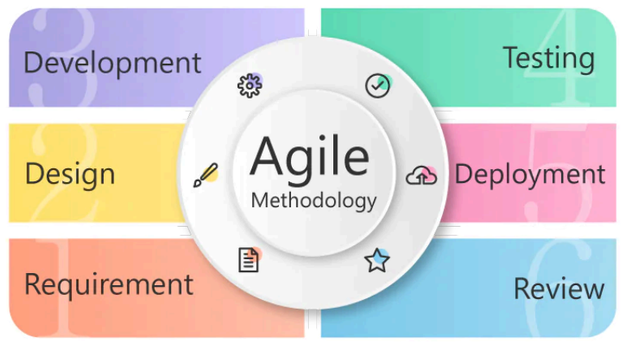
OBJECTIVES

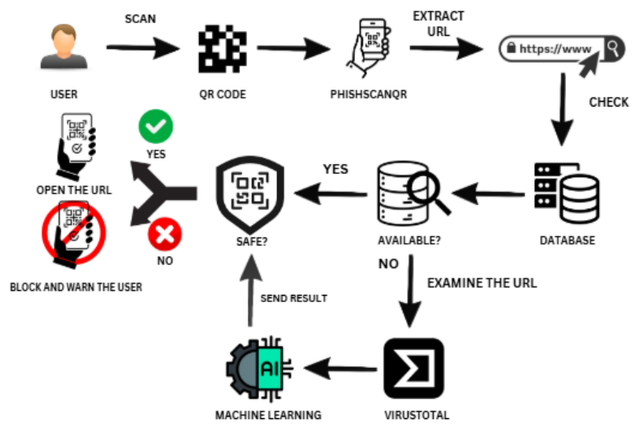
- To design a mobile application system using Android Studio that supports the scanning of QR codes for phishing detection.
- To develop a mobile application with machine learning model using the Random Forest algorithm in Google Colab using the Python language, with the VirusTotal API for determining whether QR code URLs are phishing or secure.
- To test the functionality of the mobile application based on a dataset of QR codes from Kaggle and performance metrics such as precision, recall, and F1-score.

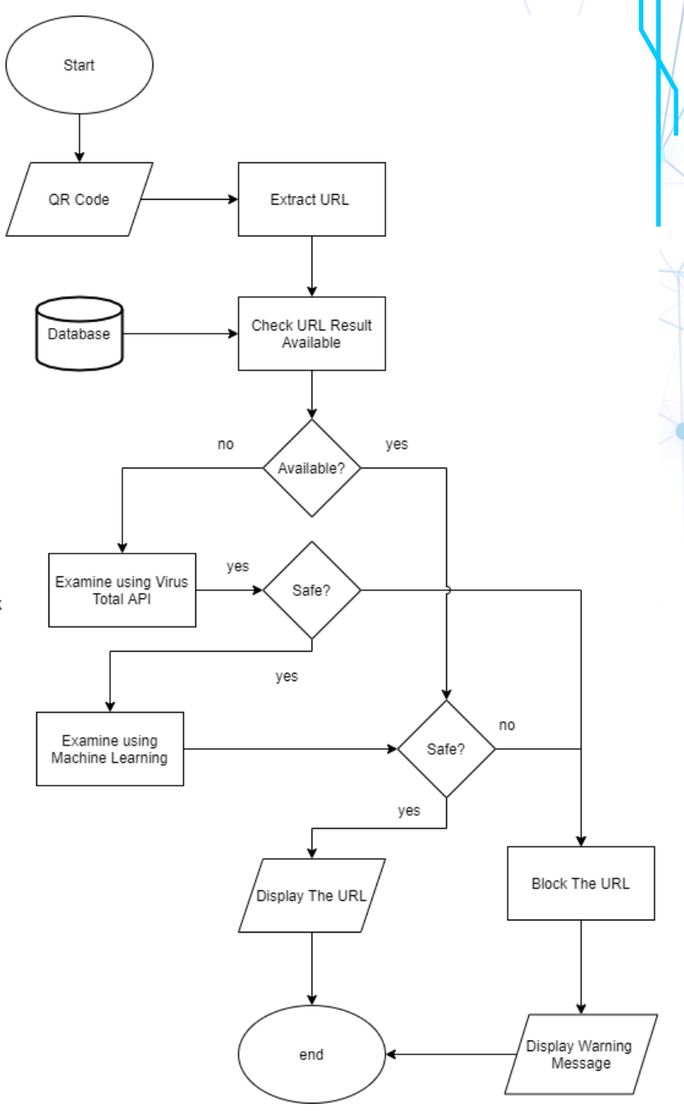
PROJECT SIGNIFICANCE

- Improving Cybersecurity Against Emerging Threats
- Creating Public Awareness on QR Code Phishing
- Encouraging the Use of AI in Mobile Security
- Developing an Accessible and Affordable Solution

METHODOLOGY



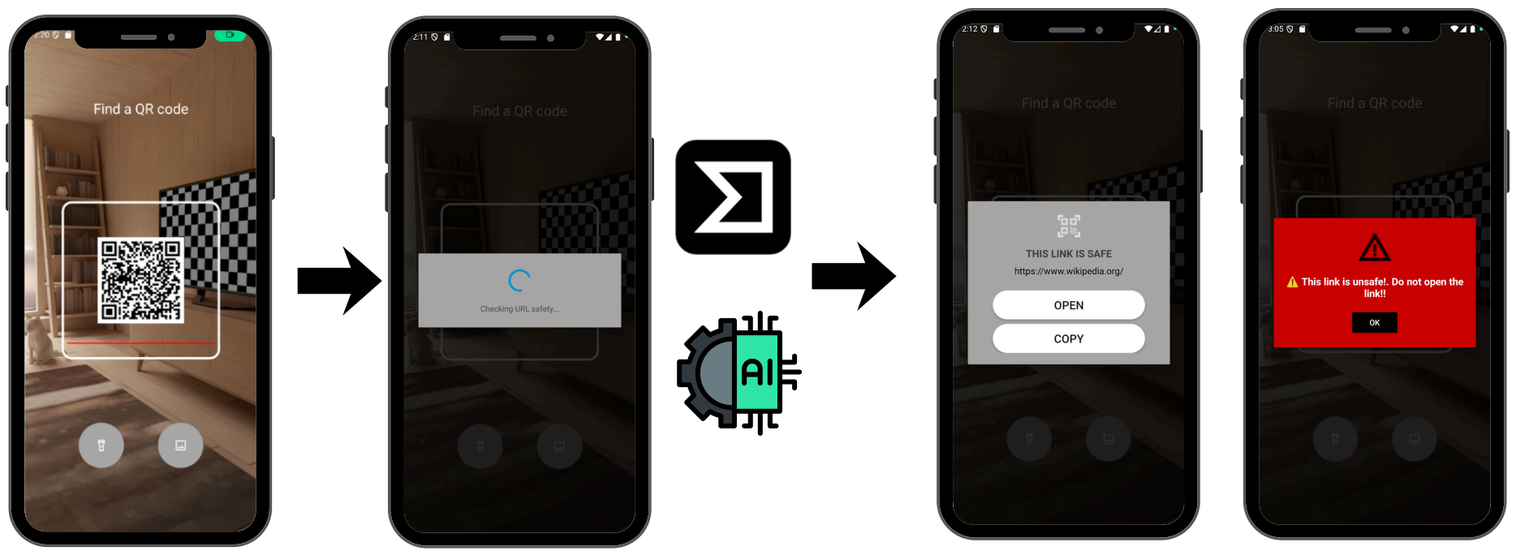




RESULTS/FINDINGS

MACHINE LEARNING MODEL ACCURACY

ACCURACY	**87%**
PRECISION	**90%**
RECALL	**84%**
F1-SCORE	**87%**
CROSS VALIDATION	**86%**



IMPACT ON SOCIETY

- **Protection Against Financial Fraud:** The system directly reduces the risk of financial loss for individuals by detecting and blocking QR codes linked to fake banking sites
- **Enhancement of Cybersecurity Awareness:** By providing explicit "Warning" and "Safe" feedback, the application serves an educational role, training users to be more cautious
- **Safeguarding Vulnerable Demographics**: The application acts as a digital guardian for non-technical users, such as the elderly or children, who may not possess the cybersecurity knowledge
- **Reduction of Cybercrime Success Rates:** Widespread adoption of such detection tools forces attackers to expend more resources to bypass security

REFERENCES

Banoth, S., Chandragiri, B., Ramadugala, P., Oraganti, H., & Konapakula, J. (2025). Detecting phishing URL using random forest classifier. World Journal of Advanced Research and Reviews, 25(2), 762–769. https://doi.org/10.30574/wjarr.2025.25.2.0360

Barron, I. R., & Sharma, G. (2025). Quashing quishing attacks using Self-Authenticating Dual-Modulated QR codes. IEEE Security & Privacy, 2–10. https://doi.org/10.1109/msec.2025.3530487

Subairu, S., Alhassan, J., Abdulhamid, S., & Ojeniyi, J. (n.d.). [A Review of Detection Methodologies for Quick Response code Phishing Attacks]. A Review of Detection Methodologies for Quick Response Code Phishing Attacks.

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